SYNTHETIC DATA · PORTFOLIO PROJECT

Customer Segmentation & Retention Analytics

A fictional retail-banking book of 4,000 customers, segmented on relationship value × engagement, with retention warning signals. Every record, outcome and figure is synthetic.

Snapshot 30 Sep 2026 · values in USD
Author: Ahsan Anas · built with AI assistance

Region: All regions | Age band: All ages | Main bank: All

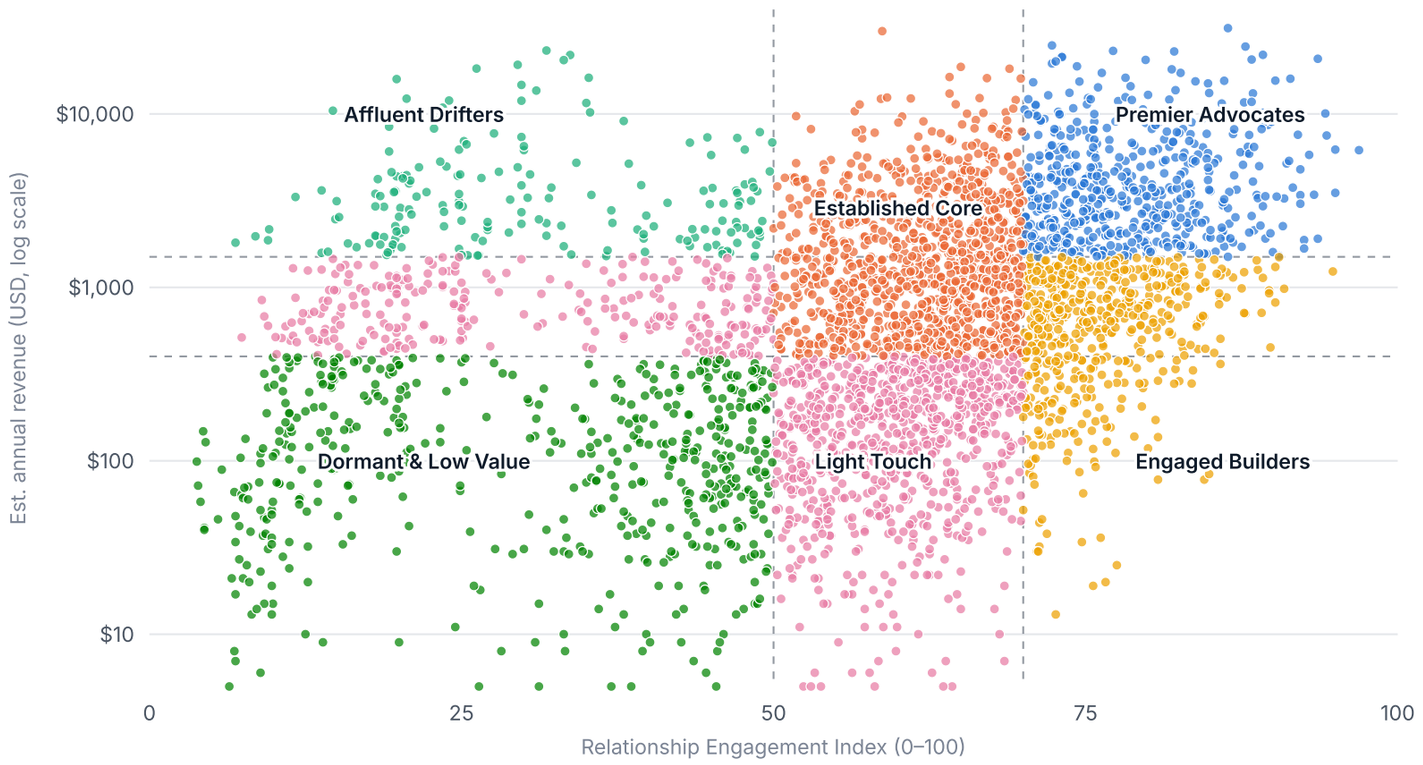
Segments: Premier Advocates | Established Core | Affluent Drifters | Engaged Builders | Light Touch | Dormant & Low Value

Customers	Est. annual revenue	Median revenue / customer	Avg engagement index	Transacted in last 30 days	Attrited next 6 months	Expected revenue at risk
4,000	$6.56m	$674	58.2	77.7%	9.9%	$533.1k
100.0% of the book	100.0% of total	revenue proxy (assumptions in README)	REI, 0–100	recency	synthetic outcome	model estimate

WHO THE CUSTOMERS ARE

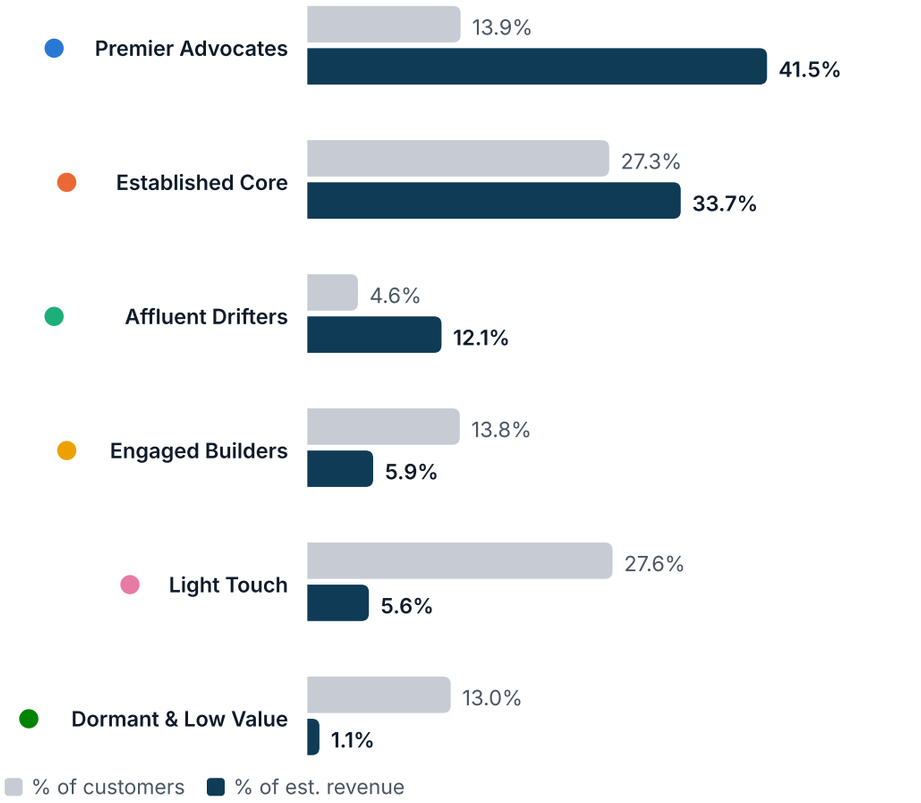
Segment map: relationship value × engagement — SYNTHETIC OBSERVATION

Each dot is one fictional customer. Dashed lines are the segment cut-offs (value USD 400 / 1,500 · engagement 50 / 70). Hover a dot for details.



Share of customers vs share of revenue — SYNTHETIC OBSERVATION

Premier Advocates are 14% of customers but 42% of est. revenue (filtered view)



What drives engagement — SYNTHETIC OBSERVATION

Average Relationship Engagement Index points by component (share of each component's maximum)

	Recency	Frequency	Digital	Adoption	Loyalty
Premier Advocates	87%	78%	56%	84%	77%
Established Core	74%	65%	44%	55%	60%
Affluent Drifters	13%	19%	14%	65%	54%
Engaged Builders	91%	79%	64%	57%	79%
Light Touch	67%	56%	40%	31%	61%
Dormant & Low Value	31%	33%	23%	18%	53%

Product adoption by segment — SYNTHETIC OBSERVATION

% of customers in each segment holding the product

	Sav.	TD	Card	Loan	Mort.	Inv.	Ins.
Premier Advocates	87%	37%	72%	27%	63%	32%	32%
Established Core	69%	21%	48%	22%	31%	15%	14%
Affluent Drifters	69%	33%	46%	11%	68%	26%	13%
Engaged Builders	76%	26%	60%	22%	7%	11%	28%
Light Touch	53%	11%	30%	7%	4%	5%	15%
Dormant & Low Value	33%	7%	15%	3%	1%	3%	11%

Segment profile table — SYNTHETIC OBSERVATION

Filtered view · medians unless stated

Segment	Cust.	Revenue	REI	Days since txn	Products	Attrition
Premier Advocates	554	$3.4k	79	5	4.5	2.2%
Established Core	1,091	$1.2k	62	12	3.2	7.8%
Affluent Drifters	183	$2.9k	31	162	3.7	28.4%
Engaged Builders	551	$687	76	3	3.3	2.7%
Light Touch	1,103	$226	53	11	2.3	10.6%
Dormant & Low Value	518	$104	31	53	1.7	22.0%

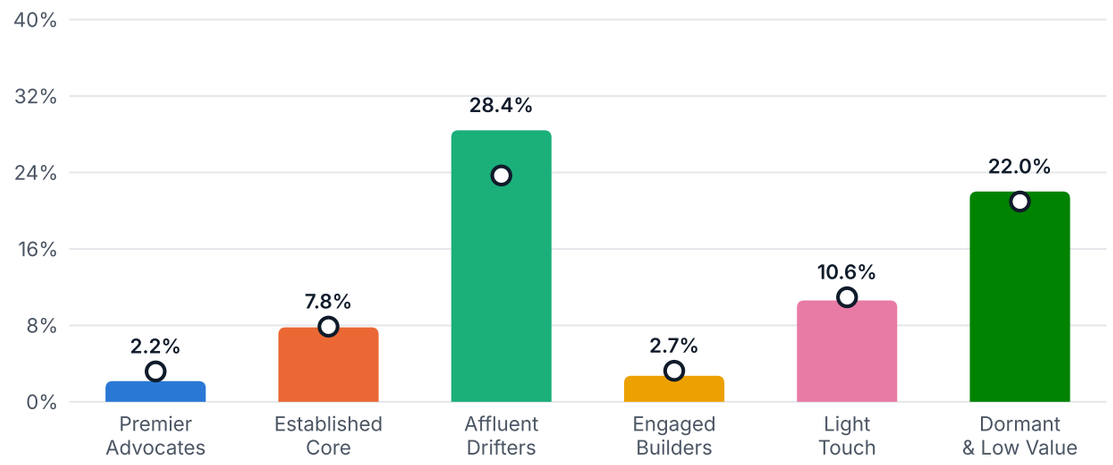
Revenue and days: medians · REI, products: averages · attrition: synthetic 6-month outcome

Source: data/processed/customers_scored.csv · pipeline: python -m src.run_all

RETENTION

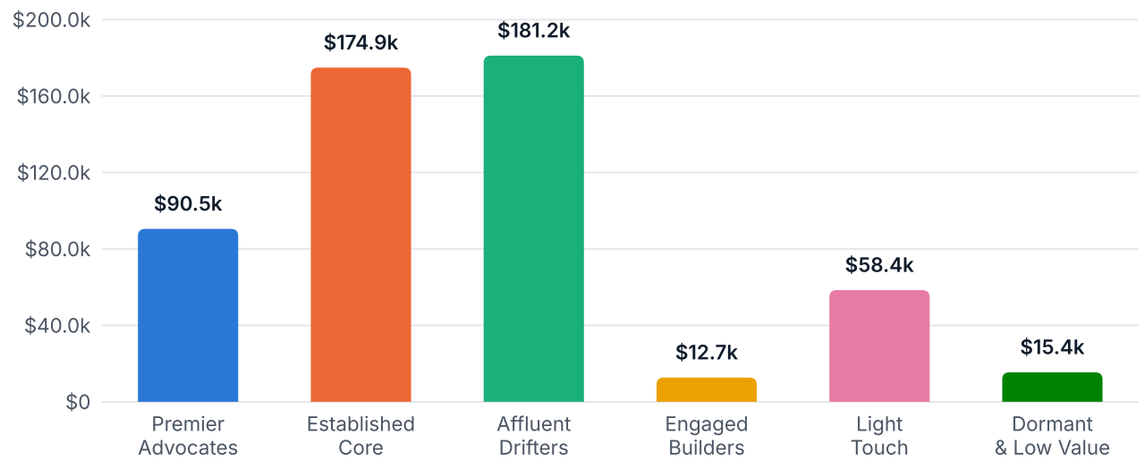
Attrition by segment SYNTHETIC OUTCOME

Bars: share that attrited in the following 6 months (synthetic label) · dot: average model-estimated risk



Where the expected revenue at risk sits MODEL ESTIMATE

Est. annual revenue × model attrition probability, summed by segment (filtered view)



Which warning signals matter? SYNTHETIC OBSERVATION

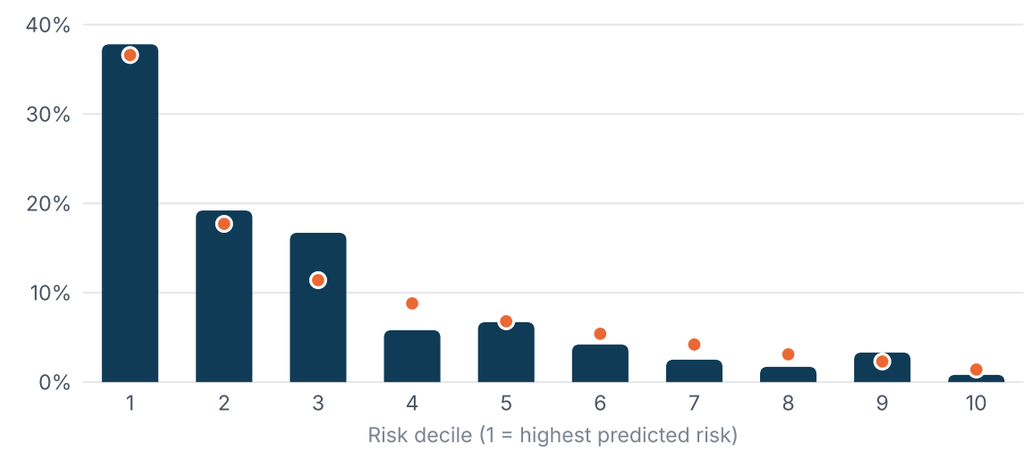
Attrition rate when the signal is present vs absent · lift = rate when present ÷ book rate (9.9%)

Signal	Flagged	Attrition if present	if absent	Lift
App registered but unused 30+ days	296	30.1%	8.3%	3.0×
No transaction for 45+ days	671	24.9%	6.8%	2.5×
2+ complaints in 12 months	131	21.4%	9.5%	2.2×
Deposits down 15%+ over 6 months	406	20.7%	8.7%	2.1×
Current account only	414	15.2%	9.3%	1.5×
Not main bank	2,121	14.9%	4.2%	1.5×
Card utilisation 85%+	5	0.0%	9.9%	0.0×

Not every intuitive signal helps: high card utilisation flags only a handful of customers and shows no extra attrition in this synthetic book.

Attrition model: does the ranking work? MODEL ON SYNTHETIC DATA

Logistic regression, 30% held-out test set · AUC 0.81 · top 10% of scores capture 3.8× the average rate



The outcome was simulated from the same hidden traits as the inputs, so this shows the method works – not that it would perform this well on real customers.

OPPORTUNITIES, ROBUSTNESS AND DATA QUALITY

Hypothetical interventions HYPOTHETICAL – ASSUMPTION-DRIVEN

Illustrative sizing only. Each number depends entirely on the stated assumption; a holdout pilot would be needed to measure real effects.

Affluent Drifters: Relationship-manager re-engagement call + needs review

$45.3k

Expected annual revenue protected (USD)

Assumption: Outreach cuts their attrition probability by 25% (relative) · 183 customers

Engaged Builders: Targeted cross-sell of one savings or card product

$9.9k

Incremental annual revenue (USD)

Assumption: 10% take up one product worth USD 180 a year · 551 customers

Light Touch: Digital activation (app onboarding, e-statements, bill-pay set-up)

4.8

Customers retained (expected)

Assumption: 15% become regular app users; their model risk falls to that of current regular app users in the segment · 1,103 customers

Pilot design: test before scaling PROPOSED

Random 50/50 split · smallest change detectable at 95% confidence / 80% power (indicative)

Cell	Treat / holdout	Baseline	Detectable change
Affluent Drifters RM outreach	91 / 91	28.4%	±18.7 pts
Light Touch digital activation	548 / 547	10.6%	±5.2 pts
Engaged Builders cross-sell	273 / 273	5.0%	±5.2 pts

Baselines are synthetic attrition rates (cross-sell: assumed 5% take-up). Small cells such as Affluent Drifters can only detect large effects – pool or extend the pilot.

How stable are the segments? SENSITIVITY TEST

Customers keeping the same segment under alternative weights and cut-offs

Scenario	Same segment	ARI
Recency-led (40/15/15/15/15)	89.7%	0.76
Adoption-led (25/15/15/30/15)	86.8%	0.71
Equal weights (20 each)	88.6%	0.74
No loyalty (35/25/15/25/0)	88.0%	0.72
Digital-led (25/15/30/15/15)	84.4%	0.65
Value bands -20%	92.9%	0.84
Value bands +20%	93.9%	0.85
Engagement bands -5 pts	80.1%	0.57
Engagement bands +5 pts	79.1%	0.57

k-means on the same behaviours finds weak natural structure (silhouette 0.20–0.22) and agrees only loosely with the rule-based segments (ARI 0.18). The rules were kept because each segment maps to a clear business action.

Data quality: what was fixed SYNTHETIC ISSUES

4,085 raw rows → 4,000 customers · 14 checks · nothing imputed

Check	Rows	Action
Exact duplicate rows	40	Removed
Re-ingested older snapshots	45	Kept newest snapshot
Region coding variants	70	Standardised
Employment text case / whitespace	55	Trimmed, re-cased
Mixed date formats (last_login_date)	140	Parsed each format
Transaction dated after the snapshot	7	Set unknown, flagged
Impossible age	9	Set missing
Sentinel tenure	24	Set missing
Income recorded in thousands	36	×1,000 unit fix
Income missing	150	Left missing (not imputed)
Impossible card utilisation	14	Set missing
Card holder without a limit	11	Left missing, flagged
Stale product count	60	Recalculated from flags
Negative savings balance	18	0 if no product, else missing